HTML

Introduction

- Hyper Text Markup Language
- .html or .htm
- Construction
  - Tags: Start & End
  - Element: Simple & Complex
  - Entity reference
  - Attribute

Sample HTML Page

```
<!DOCTYPE HTML>
<html>
<head>
  <title>Simple Tags Demo</title>
<!-- <title> Old Title </title> -->
</head>
<body>
<a href="http://www.pgcpac.cdac/" title=
    “PGCPAC">This is a link.</a>
<br />
<img src="logo.gif" alt="logo" />
<br />
<strong>Bold</strong> and <em>italic</em> text.
</body>
</html>
```

Block and Inline Elements

- Block elements add a line break before and after them
  - `<div>` is a block element
  - Other block elements are `<table>`, `<hr>`, headings, lists, `<p>` and etc.
- Inline elements don't break the text before and after them
  - `<span>` is an inline element
  - Most HTML elements are inline, e.g. `<a>`

Headings and Paragraphs

- Heading Tags (h1 – h6)

```
<h1>Heading 1</h1>
<h2>Sub heading 2</h2>
<h3>Sub heading 3</h3>
```

- Paragraph Tags

```
<p>This is my first paragraph</p>
<p>This is my second paragraph</p>
```

- Sections: `div` and `span`

```
<div style="background: skyblue;">
   This is a div</div>
```

Text Formatting

- Text formatting tags modify the text between the opening tag and the closing tag
  - Ex. `<b>Hello</b>` makes “Hello” bold

	bold
	italicized
<u></u>	<u>underlined</u>
	Sample^{superscript}
	Sample$_{\text{subscript}}$
	strong
	emphasized
<pre></pre>	Preformatted text
<blockquote></blockquote>	Quoted text block
<del></del>	Deleted text – ~~strike through~~

Important Tags

- <a> - File, URL, Section
- <img>
- <ol type="1"><li> - 1, I, i, a, A
- <ul><li> - disc, circle, square
- <dl><dt><dd> definition list, text
- <script>
- <meta name="viewport" content="width=device-width, initial-scale=1.0">
- <title>
- <style>
- <link rel="icon" href="favicon.ico"/>

Table

- <table>
- <caption>
- <tr>
- <th>
- <td>
- <thead>
- <tbody>
- <thead><tbody>
  <tfoot> (grouping)

- Attributes
  - rowspan
  - colspan
  - id
  - cellspacing
  - cellpadding

Form

- Collects user input
- <form action="page1.php" method="POST" enctype="multipart/form-data“>

  </form>
- GET vs POST
  - Input in URL
  - data length restriction
  - browser history
  - cached

Form Elements

- <fieldset><legend>
- <label><input><button>
- <textarea rows=“” cols=“”>
- <select name="sex">

```
    <option value="1">Male</option>
    <option value="2">Female</option>
    <option value="3">Transgender</option>
    <option value="4">Unknown</option>
</select>
```

Attributes: multiple, selected

Input

- Type
  - text
  - password
  - hidden
  - file
  - radio
  - checkbox
  - submit
  - reset
  - email
  - number (min, max, step)
- Type
  - date
  - time
  - range
  - color
- Attributes
  - required
  - disabled
  - readonly
  - cellpadding

Multimedia

- <video width="320" height="240" autoplay controls>

  <source src="sample.mp4" type="video/mp4"> <track src="abc.vtt">

  </video>
- <audio autoplay controls>

  <source src="sample.mp3" type="audio/mpeg">

  </audio>

HTML Special Characters

Symbol Name	HTML Entity	Symbol
Copyright Sign	©	©
Registered Trademark Sign	®	®
Trademark Sign	™	™
Less Than	<	<
Greater Than	>	>
Ampersand	&	&
Non-breaking Space		
Em Dash	—	—
Quotation Mark	"	"
Euro	€	€
British Pound	£	£
Rupee	₹	₹